List of theoretical equations which could be in the exam

This is a list of theorems and results for which the proof or parts of the proof may be asked in the exam:

1. Asymptotics for the probability to return to zero for Random Walks
2. Theorem 3.2.1
3. Theorem 3.2.3
4. Chapman-Kolmogorov equations for Markov Chains
5. Theorem 4.2.1
6. Theorem 4.3.3
7. Theorem 4.3.4
8. Theorem 4.3.5
9. Theorem 4.4.4
10. Property Π1
11. Property Π2
12. Property Π3
13. Property E4
14. Property PP1
15. Theorems 5.10.2 and 5.10.3
16. Lemma 5.13.1
17. Theorem 6.3.1
18. Theorem 6.5.2

The remaining theorems and results, such as Theorem 5.11.1 and Theorem 7.1.1, are also important, but their proofs will not be tested.